

Traiter les questions suivantes :

Question 1 (10 pts)

- A-** A person may be a coach, co-coach, a physicist, a player, a referee or supporter. A person has a name, surname, date of birth, and a gender. A supporter has an e-mail and supports a team, a coach has a certification date, a player has a nickname and he can be a field player or a goalkeeper. A player has the right to use his feet, a goalkeeper has the right to use his hands too. A player signs a contract with a team. A contract has a number, a start date, an end date, a value in dollars and a penalty for cancellation. A team has a name and a date of establishment and represents a city. For a given city, every team has a unique name. A referee may be a field referee or a line referee. A game connects two teams at a given date and at a given season. A match is followed by three referees, two line referees and one field referee. Before each game, a team must submit a list containing the names of the players and of the reserve players that will participate in the game, with their numbers,. For each game, you must store the number of touches, the number of minutes of participation for each player and the number of goals scored. A player may score many goals in a game and you must annotate the time in minutes of each goal scored. A player may obtain several yellow cards (maximum 2 for a given game) and several red cards (maximum 1 for a given game) and you must annotate in both cases the time in minutes of each card obtained. Each game has one or more sponsors. A sponsor is characterized by his number, name and the product that he publishes. Construct the corresponding class diagram.
- B-** Construct the object diagram that corresponds to the following scenario: The game number 134 of the 2004-2005 season was sponsored by Pepsi. This game is held on 12.01.2004 with field players Hassoun and Philics of Ansar team, and Halawi and Nasser Eldine of Nijmeh team. These players have scored goals respectively at times 46, 86, 90, 93. At the minute 90 the goalkeeper Al-Samad of the team Ansar takes a yellow card.
- C-** Construct the use case diagram to answer the following outcomes:
The head of the football championship wishes to have:
- A list of players per team for a given season to verify their legal situations.
- A list of coaches and co-coaches per team for a given season.
- A complete history of all games made in a given season.
Achieving these lists and the history requires the presence of a user to enter statistical data and those corresponding to the games. To check the accuracy of data inputs another user controls and modifies the data and then confirms them.
- D-** Construct the use case diagram of the following scenario:
A visitor can navigate through the site of the championship and he can register to become a supporter by completing the appropriate form. During his navigation he can consult the scorer list, the lists of championship history, the lists of teams, the championship calendar and the latest results. The supporter receives a user name and a password that allow him to identify himself and to purchase tickets online by credit card.
- E-** Construct the state diagram of the object coach in the following scenario: At the beginning of the game, the status of the coach is calm. If the referee whistles on his team, he gets angry if he was calm and he becomes calm if he was happy. If the referee whistles with his team he becomes happy if he was calm and he becomes calm if he was angry. If his team scores a goal he becomes happy while jumping. If the first half is over, the coach enters in a meeting with his players and he returns in a calm state. At the end of the second half, when the referee whistles the end of the game, the coach smiled in the case of victory and does nothing otherwise.
- F-** Construct the sequence and activity diagrams of the following scenario: A supporter demands the purchase of a ticket; the system will request a username and a password for identification. The supporter enters his username and password, the system processes them and gives a message of success or failure according to the case. In case of failure the supporter will be entitled to two other trials. If the identification is successful the system asks him to enter the desired game. If the supporter is not a supporter of one of the Two teams participating in the corresponding game or if no tickets are available for this game, the scenario ends with a regret message. Otherwise, the system asks the supporter to indicate the type of ticket among the types that are available and the supporter suggests one in his turn. The system displays the form of payment, the supporter enters his credit card number then the system makes verification with the system corresponding to the payment type. According to the result of authorization, the system indicates to the supporter the failure or the success of the operation.

Question 2 (3 pts)

- Construct an activity diagram to model the process of a product order. The process involves the following actors:
Customer: orders a product and pays the invoice
Cash desk: to cash the money from the customer
Vendor: responsible to treat and to invoice the customer's order.
Warehouse responsible: responsible for preparing and shipping the items corresponding to the order.

Question 3 (7 pts)

In this exercise, we wish to develop the state and interaction model of a telephone.

- a)** Draw the sequence diagram corresponding to the case where a user (the caller) calls another user (the called person). Assume that the case is normal (the caller dials a valid number and the called person is not busy and answers). We assume that the entities of the scenario are the caller, the telephone of the caller, the telephone line, the telephone of the called person and finally the called person. Assume that the caller hangs up first.
(This question helps to identify the different events sent and received by telephone from or to the telephone line and the user, and to identify that the device is called or calling)
 - b)** Complete the previous scenario with the case where the called person is busy and the caller dials a wrong number.
 - c)** Construct a collaboration diagram around the class 'Telephone' summarizing all the events identified by the previous questions.
 - d)** Construct the state diagram of the telephone. Tip: You can have the macro-state 'Hung' with the sub-states 'Silent' and 'Ringing' and the macro state 'Pick up' with the sub-states 'BaseTone', 'OtherSearch', 'WrongNumber', 'OtherBusy', 'OtherRing', 'OtherHung' and 'communication'.
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